

Track 26: Computer Vision

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Python → OpenCV → Image Processing → Deep Learning → CNNs → Object Detection → Deployment

Related: [Track 5 — AI/ML](#) | [Track 21 — Raspberry Pi](#) | [Track 24 — Drone Programming](#)

Track Overview

Who this is for: AI/ML engineers, robotics developers, and developers building visual intelligence systems.

Prerequisites: Python; basic linear algebra helpful; [Track 5](#) ML basics for 3-month+ tiers.

Tools: Python, OpenCV, NumPy, PyTorch/TensorFlow, YOLO, MediaPipe, Roboflow, Jupyter, ONNX/TensorRT (6-month+).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	CV awareness	Face detection + filter app
1 Month	80–100	Junior CV developer	Object counter app
3 Months	250–300	Internship-ready	Custom object detection model
6 Months	500–600	CV engineer entry	Real-time detection deployment
9 Months	750–900	Advanced CV engineer	Multi-model vision platform

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Images as arrays, OpenCV basics, read/write/display	Image manipulation toolkit
D3–4	Filters, edge detection, contours	Edge detection app
D5–6	Haar cascades, face/eye detection	Face detection webcam app
D7–8	Video processing, background subtraction	Motion detection system
D9–10	Capstone integration	Live webcam vision app

1-Month Foundation

Week	Topics	Project
W1	OpenCV deep dive, morphological ops, histograms	Image enhancement pipeline
W2	Feature detection (SIFT/ORB intro), matching	Document scanner
W3	CNN basics, transfer learning (MobileNet)	Image classifier
W4	YOLO/SSD intro, inference	Object counter/detector app

3-Month Job Ready

Month	Topics	Projects
M1	OpenCV advanced, camera calibration	AR overlay project
M2	Dataset labeling (Roboflow), training YOLO	Custom detector (3 classes)
M3	Model optimization, deployment, capstone	Deployed CV application

Capstone: Train on custom dataset; >80% mAP; real-time inference demo; Flask/Streamlit UI.

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	Semantic segmentation, pose estimation (MediaPipe)	People counter with pose
M5	Edge deployment (Pi/Jetson), ONNX	Edge inference pipeline
M6	Capstone	Industrial inspection / surveillance CV system

9-Month Advanced

Capstones: Multi-camera vision analytics platform + autonomous vision module (drone/robot) |
Level: L5

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	OpenCV, CNN, detection metrics
Lab Completion	40%	Code runs; models train/infer
Project	30%	Accuracy, speed, deployment
Portfolio	10%	GitHub, demo videos, model cards

Appendix: Mentor Notes

GPU recommended for training labs (Colab free tier acceptable). Cross-track with [Track 24](#) for aerial CV.